Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

beamer [secheader]beamerthemeBoadilla tcolorbox tikz mwe
siunitx hyperref tikz tcolorbox []beamercolorthemespruce colortbl
xcolor graphicx positioning, arrows.meta color epsfig ifpdf
[latin1]inputenc multimedia
latexsym amsmath,amsthm,amssymb mathrsfs
listings xcolor
language=Python, basicstyle=, keywordstyle=,
commentstyle=, numberstyle=, stringstyle=, breaklines=true,
breakatwhitespace=true, frame=single, showstringspaces=false, captionpos=b,

Your Title

Kainat Naeem

Your Department Name
Your Faculty Name
Your Institute Name

19 Pebruari 2025

Outline

Introduction

Section 2 Title

Section 3 Title

Section 4 Title

Introduction

Title

Write definition

Examples:

- Point 1.
- Point 2.
- Point 3.
- Point 4.

Blocks

The blocks are shown below

Main Block

Content of a main block

Example

Content of an example block

Note

Content of an Important note block

Subheading 1

- Point 1.
- Recent research [1]
- Zhang[2]

[1] reference

[2] reference 2

Section 2 Title

Title

yellow!20 yellow**Block 1**	green**Block 2**	yellow**Block 3**	green**Block**
1	2	3	2004
1	2	3	2004
1	2	3	2004

Tabel: Captionfor table

- Point 1 [3]
- Point 2 [4]

[3] Refrence 1
[4] Refrence 2

Title of Slide

```
    Here you write your MATLAB code
it will appear as it is due to verbatim command
```

- C, d
- After `cvx_end`
- Variable z

Title of Slide

```
using Convex

% Comment
z = Variable(q)

% Add Comment
solution = z.value
```

- C, d
- After `solve!`,

Syntax

[language=Python,basicstyle = ,] `import library`
`z = cp.Variable(variablename)`
$\texttt{opt}_v al = probl.solve()solution = z.value$

- C, d
- solve!

Section 3 Title

Blocks

The blocks are shown below

Main Block

Content of a main block

Example

Content of an example block

Note

Content of an Important note block

Two columns

value of column one

$$a^2 = b^2 + c^2 \qquad (1)$$

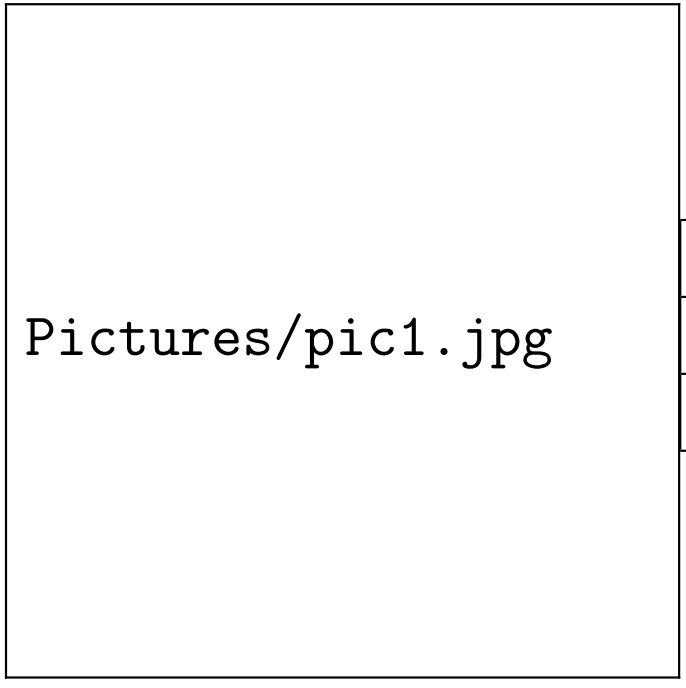
value of column two

$$x = 2y + z \qquad (2)$$

Section 4 Title

Insert Title Of Slide

- Explanation to pic and table .



blue!25 **Block 1**	**Block 2**	**Block 3**
1	5	20.0
2	-4.2	14.2

- Your Explanation Here.

Conculsion

- Your Conculsions.

Pictures/thankyou.png